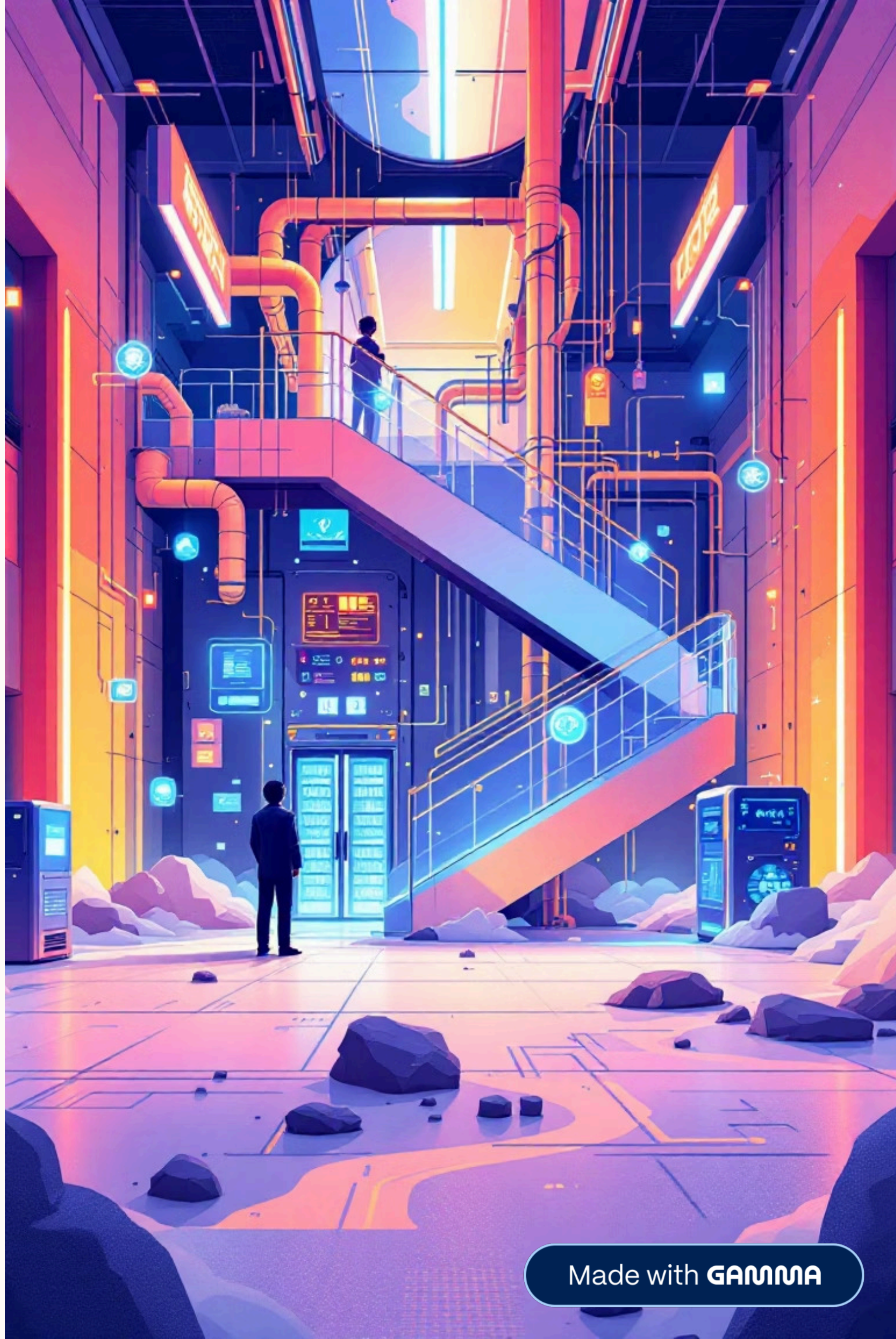
AI-Powered Smart Cold Chain: Predictive Spoilage Monitoring System

An AI + IoT solution for real-time monitoring, predictive analytics, and spoilage prevention.

Presented by: Tanishq Kolhatkar

Sabudh Foundation - AIOT Internship



The Cold Chain Challenge: A Costly Problem

1. **Unmonitored cold chains lead to significant losses, impacting businesses and consumers alike. The lack of real-time insights and predictive capabilities creates vulnerabilities at every step.**
2. **Frequent spoilage due to temperature/humidity fluctuations**
3. **No real-time visibility across cold storage/transport units**
4. **Manual monitoring fails to detect early spoilage**
5. **Industry losses due to unmonitored cold chain failures**
6. **Lack of predictive analytics or alerting mechanisms**

Global food waste due to cold chain inefficiencies is estimated at over 1.6 billion tons annually, costing billions of dollars.

Our AI-Powered Solution: Predictive Spoilage Monitoring

Leveraging the power of IoT and AI, our system provides continuous monitoring and predictive analytics to prevent spoilage before it occurs, ensuring product integrity from farm to fork.

IoT Sensor Data

Continuous monitoring of temperature, pressure, fill level, and moisture.

ThingSpeak Cloud

Secure data transmission and storage in the cloud.

MATLAB Analytics

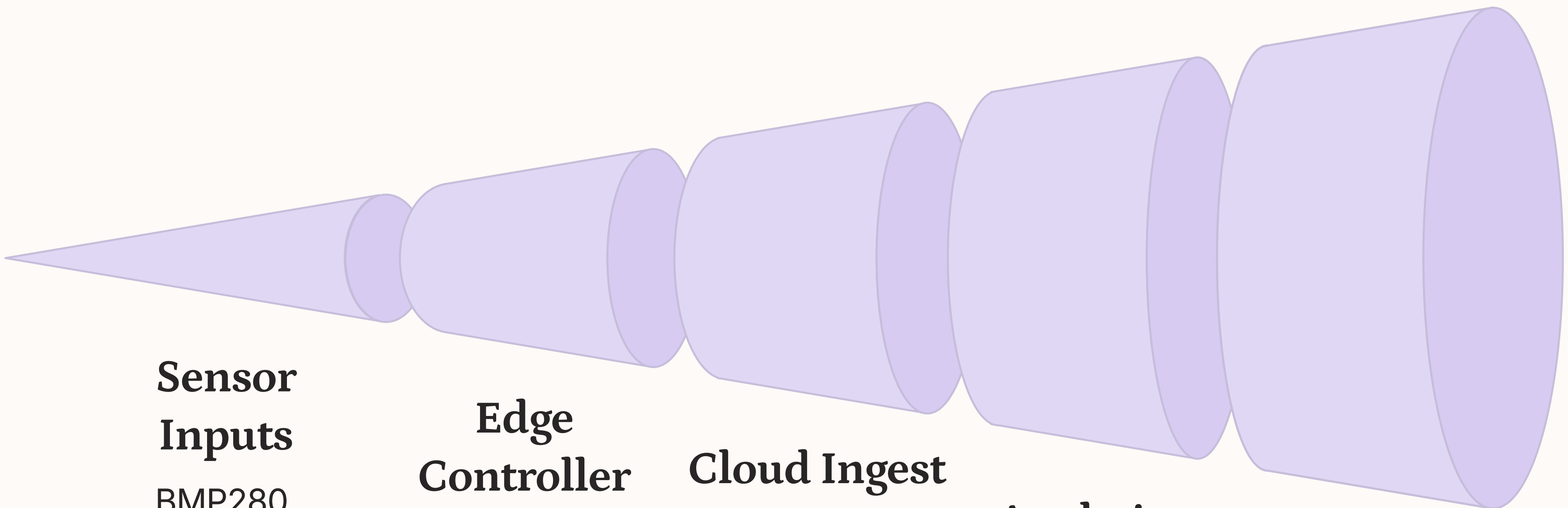
Anomaly detection and predictive spoilage modeling using AI.

Flutter Mobile App

Real-time dashboard, alerts, and notifications for users.

Comprehensive System Architecture

Our robust technical architecture integrates diverse components, forming a seamless, intelligent monitoring ecosystem.



Sensor Inputs

BMP280, Ultrasonic, Capacitive Moisture

Edge Controller

ESP32 firmware with Wi-Fi

Cloud Ingest

ThingSpeak channels & fields

Analytics

MATLAB thresholds & anomaly detection

Delivery & Alerts

REST API, Flutter app, notifications

End-to-End Platform Workflow

From data collection to predictive insights, our system provides a continuous, automated flow of information to ensure optimal cold chain management.

1

Sensor Data Collection

Environmental data (temp, humidity, pressure, moisture) is continuously collected.

2

ESP32 Preprocessing & Publish

Data is preprocessed by ESP32 and securely published to ThingSpeak.

3

ThingSpeak Storage & Trigger

Data is stored, and MATLAB scripts are automatically triggered for analysis.

4

MATLAB Prediction & Scoring

AI models predict spoilage risk and calculate anomaly scores.

5

Flutter App Data Fetch

The mobile app fetches updated values and predictions via REST API.

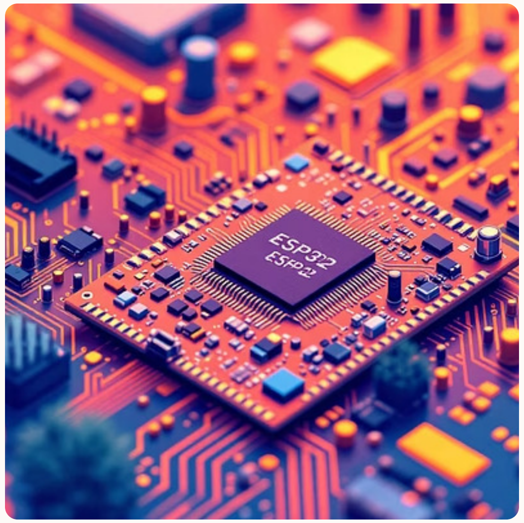
6

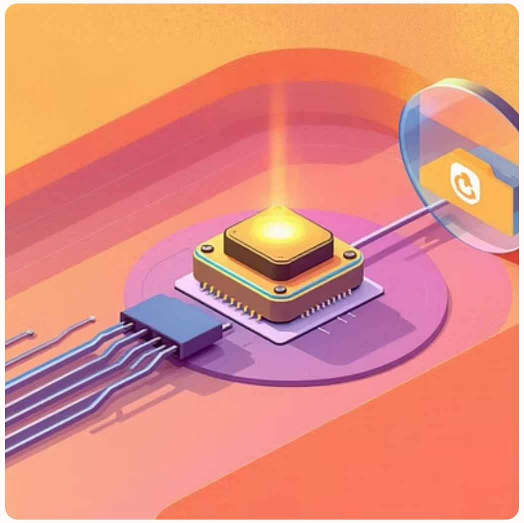
User Alerts & Insights

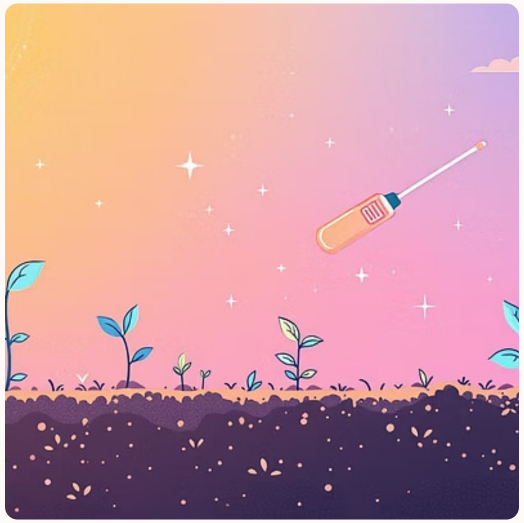
Users receive real-time alerts and comprehensive dashboard insights.

Core Hardware Components

The foundation of our system lies in reliable, accurate, and energy-efficient hardware, designed for seamless integration within various cold chain environments.







ESP32

Low-power, dual-core Wi-Fi & Bluetooth microcontroller for data acquisition and transmission.

BMP280

High-precision sensor for accurate temperature and barometric pressure readings.

Ultrasonic Sensor

Detects door-open events and monitors fill levels with sound waves.

Capacitive Moisture Sensor

Monitors moisture levels, crucial for preventing condensation and spoilage.

Power Unit

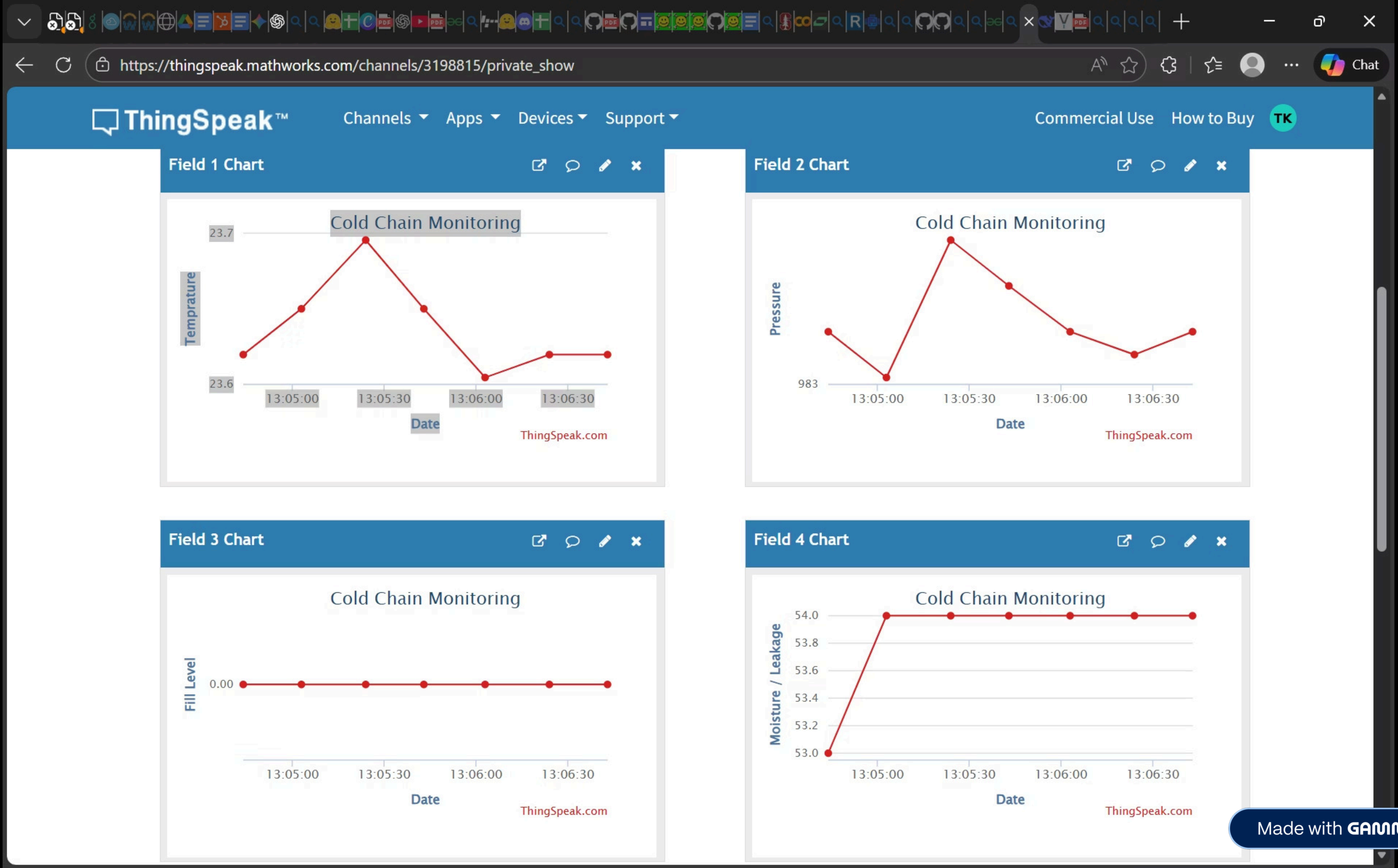
Ensures continuous operation with efficient and long-lasting power supply.

Cloud & Analytics: The Brain of the System

Our cloud-based analytics layer processes raw sensor data, applies intelligent algorithms, and predicts spoilage risks with unparalleled accuracy.

- **ThingSpeak Channel Mapping:** Organized data streams for each sensor and cold chain unit.
- **Data Publishing Frequency:** Configurable intervals for real-time updates and efficient data transfer.
- **MATLAB Threshold Rules:** Dynamic rules for identifying critical deviations in environmental parameters.
- **Time-Series Anomaly Detection:** AI algorithms identify unusual patterns indicative of potential spoilage.
- **Spoilage Risk Scoring Model:** Proprietary model generates a quantifiable risk score for proactive intervention.

ThingSpeak, an IoT analytics platform, seamlessly integrates with MATLAB, empowering our AI models to detect anomalies and predict spoilage events. This powerful combination transforms raw data into actionable intelligence.


https://thingspeak.mathworks.com/channels/3198815/private_show
ThingSpeak™
Channels
Apps
Devices
Support
Commercial Use
How to Buy
TK
Field 1 Chart
Cold Chain Monitoring
Temprature
23.7
23.6
13:05:00
13:05:30
13:06:00
13:06:30
Date
ThingSpeak.com
Field 2 Chart
Cold Chain Monitoring
Pressure
983
13:05:00
13:05:30
13:06:00
13:06:30
Date
ThingSpeak.com
Field 3 Chart
Cold Chain Monitoring
Fill Level
0.00
13:05:00
13:05:30
13:06:00
13:06:30
Date
ThingSpeak.com
Field 4 Chart
Cold Chain Monitoring
Moisture / Leakage
54.0
53.8
53.6
53.4
53.2
53.0
13:05:00
13:05:30
13:06:00
13:06:30
Date
ThingSpeak.com

Intuitive Mobile App: Flutter Power

Our Flutter-built mobile application provides users with an accessible, real-time window into their cold chain operations, delivering critical insights and alerts directly to their fingertips.

→ Real-time Dashboards

Visualize key metrics like temperature, humidity, and pressure in an instant.

→ API Data Fetching

Seamlessly pulls data and predictions from ThingSpeak via a robust REST API.

→ Spoilage Prediction Visualization

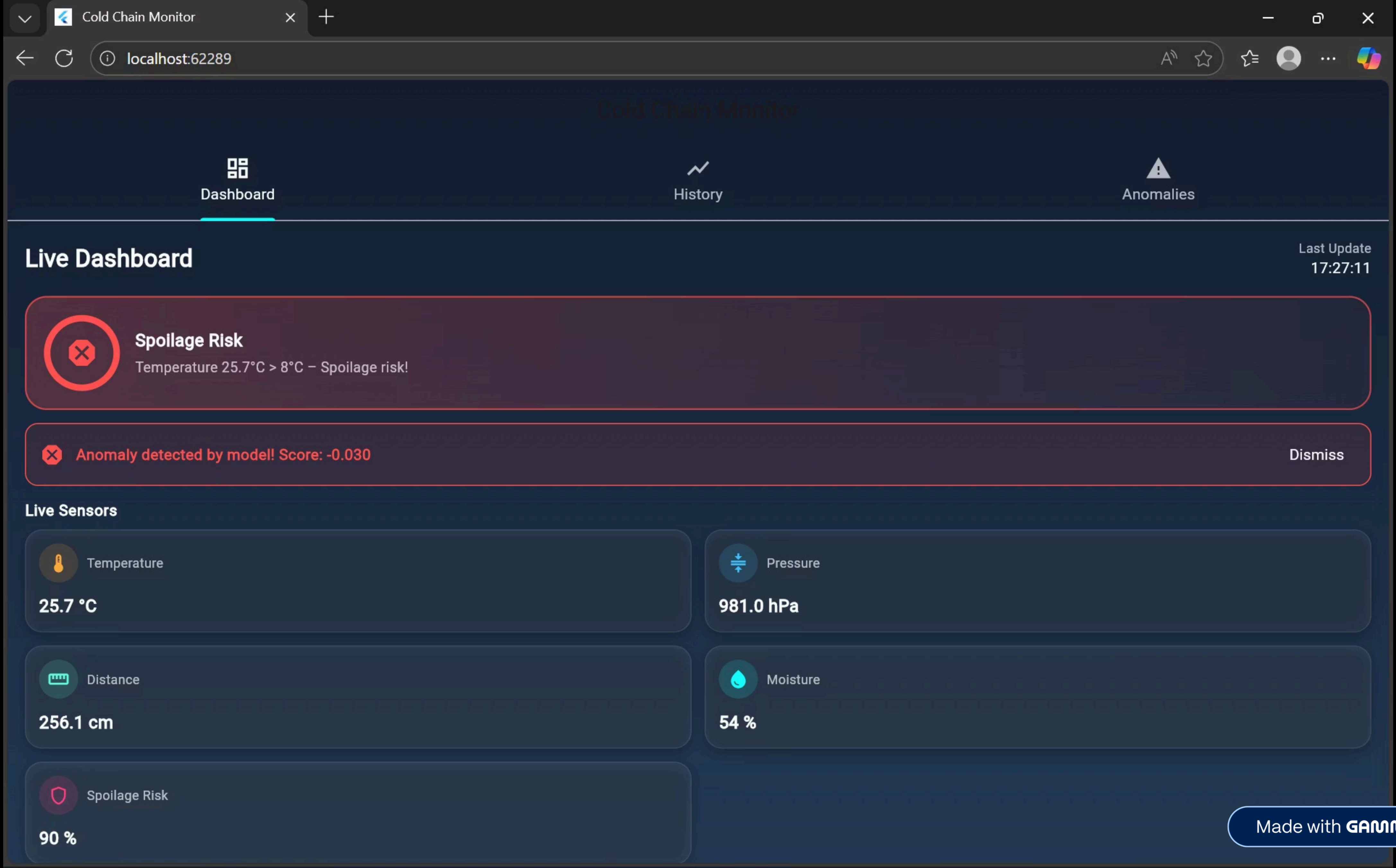
Clear, actionable display of spoilage risk scores and historical trends.

→ Push Notifications

Instant alerts for critical events, ensuring immediate response.

→ Door-Open Detection

Monitors and alerts for unauthorized or prolonged door openings.


Cold Chain Monitor
localhost:62289
Dashboard
History
Anomalies
Live Dashboard
Last Update
17:27:11
Spoilage Risk
Temperature 25.7°C > 8°C – Spoilage risk!
Anomaly detected by model! Score: -0.030
Dismiss
Live Sensors
Temperature
25.7 °C
Pressure
981.0 hPa
Distance
256.1 cm
Moisture
54 %
Spoilage Risk
90 %

Transformative Industry Impact & Use Cases

Our AI-powered cold chain solution is set to revolutionize industries by drastically reducing waste, enhancing safety, and building unprecedented trust across the supply chain.

Pharmaceuticals

Ensuring vaccine and drug efficacy with precise temperature control.

Dairy & Perishables

Minimizing spoilage of milk, cheese, and other delicate goods.

Meat & Seafood

Guaranteeing freshness and safety for highly sensitive products.

Logistics & Transport

Real-time monitoring during transit for uninterrupted cold chain integrity.

Reduces Spoilage Losses Dramatically

Improves Supply Chain Reliability

Early Warnings Prevent Accidents

Increases Customer Trust

The Future is Now: Smart, Predictive, Sustainable

Embrace the next generation of cold chain management. Our AI-powered system provides the intelligence and foresight needed to protect valuable goods, optimize operations, and foster a more sustainable future.

Thank You